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Medical waste sorting machine development with IoT and YOLO model utilization

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Abstract

The Internet of Things (IoT) application has been seen across various sectors to improve management practices. In medical waste management post-pandemic, the integration of IoT and artificial intelligence (AI) has notably enhanced sorting methods. The increased medical waste has brought about significant environmental challenges, impacting community health, land, and ocean ecosystems. We aim to develop a medical waste sorting system integrated with the You Only Look Once model (YOLO) and IoT for monitoring purposes. We can observe how effective the YOLO model is in sorting applications. This work trained and compared YOLO models from YOLO v5 to YOLO v9. YOLO v8 outperformed, and we implemented it into the sorting system. The mechanical, software, and programming elements were combined to develop the sorting prototype. From the training results, YOLO v8 achieved 98% mean average precision (mAP), 0.958 and 0.963 precision and recall, respectively. Other than that, the sorting evaluation was done during the final testing, with 93.75% accuracy. Further results were explained in the paper. The developed sorting prototype detected, classified, and sorted the medical waste, including facemasks, gloves, syringes, and sharp waste, based on the YOLO model. Further development is essential to improve the system in many aspects. Therefore, this system can be implemented practically in actual medical waste management. This work is a move that aligns with the third Sustainable Development Goal, where we focused on an automated medical waste system, ensuring sustainable health and Good Health and Well-being (SDG3).

Keywords: Artificial intelligence, Deep-learning, YOLO, IoT, SDG 3

Introduction

It is essential to categorize medical waste into different classes to prevent accidents and infections, as it may contain contaminated and infectious materials that can endanger waste handlers, such as blood and body fluids. Even though there are guidelines for managing medical and common waste, there is a risk of waste misplacement and mixing during the initial sorting process. Medical waste could harbor infectious agents such as viruses and bacteria. If it gets mixed with non-medical waste, waste handlers may inadvertently come into contact with contaminated waste, increasing the risk of infection

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and exposure to serious diseases like COVID-19 and influenza. Hidden sharp objects like needles, broken ampoules, and surgical blades in mixed waste can cause injuries to waste handlers, further elevating the health risk.

Problems arise from the exponential increase in medical waste and traditional waste sorting in the medical sector. Waste is misplaced and mixed, and manual sorting takes time. Sharp waste is also misplaced, causing accidents, infections, and virus transmission. The current disposal method ends up in landfills, which can potentially be recycled. Effective sorting at the initial phase facilitates recycling, finally helping reduce pollution to land and the ocean.

The current method for managing waste involves manual sorting. Researchers focus on developing and testing autonomous sorting technologies, utilizing various mechanisms and programming. While most efforts are concentrated on detecting and sorting general and household waste using algorithms, our research is specifically geared toward creating a machine for sorting medical waste. This machine will be capable of identifying and sorting items like facemasks, syringes, gloves, and sharp objects. Manual sorting often results in improperly disposing of sharp and medical waste with the general waste users. An automatic sorting machine can help by automatically detecting and classifying each piece of waste, thus preventing misclassification and reducing the time spent on waste management.

The medical waste sorting machines can also facilitate the waste management process by reducing the time of segregation, disinfection, and transportation to the landfill [1]. Besides avoiding health risks, medical waste can be sorted automatically and transported to the recycling center, reducing pollution. Physical and chemical valorization can be performed using the collected waste, especially the facemask. The process produced new materials such as biofuel, thickening agents, and construction materials [2]. Figure 1 displays the process of recycling medical waste after the waste has been sorted and collected. Classifying medical waste to avoid mixing accurately is critical, delaying the recycling process and facilitating recycling management at the initial stage in the medical sector and the public.

From another perspective, the research focused on a decision-making system to select the site for solid waste management based on the evaluation matrix. From the evaluation matrix and the site assessment, the high authority can monitor the information

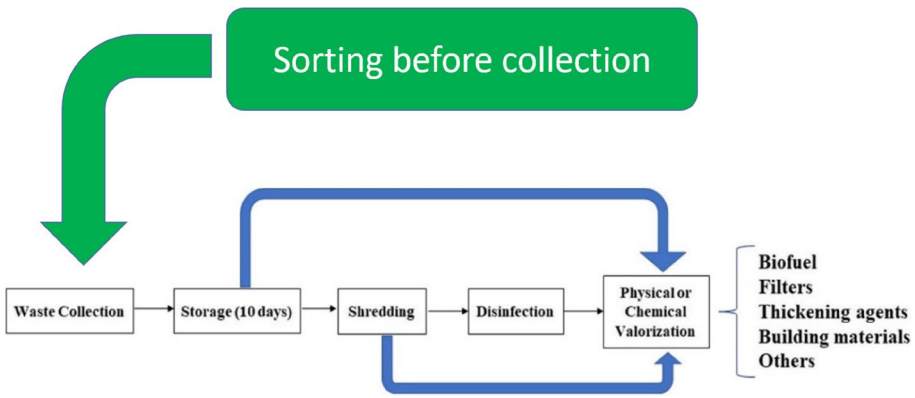


Fig. 1 Recycling process [2]

in a broader view of managing waste disposal based on factors such as environmental impact, cost, air quality index, and locality distance. The management can control the direction of large-scale waste disposal for incineration, recycling, and landfill [3]. In the medical sector, the segregated waste can be split and transported to different disposal points based on the purpose, following the above site assessment.

Our work focused on a phase in Fig. 1, initial waste disposal and sorting, before storage, treatment, and transportation. The aim is to automate the initial sorting system to replace manual disposal by the medical staff. Certain elements must be considered and emphasized when developing and implementing a sorting machine in alignment with the post-pandemic situation. The autonomous sorting system should minimize human contact with potentially hazardous waste by ensuring accurate sorting and reducing the risk of exposure to infectious materials such as medical and sharp waste. The system must adhere to existing environmental and health standards to ensure the proper and safe sorting, disposal, and recycling processes.

The YOLO model is an image training model capable of predicting multiple objects within a single image or every video frame. YOLO represents the latest advancement in Deep Learning techniques, enabling real-time object detection for identifying objects within images, videos, or camera frames. This model suits our work and automated medical waste sorting development since it can detect objects through a camera. Initially developed by Joseph Redmon, YOLO continually evolves through the efforts of other researchers who have created their versions of YOLO [4].

YOLO v5, developed by Ultralytics and released in May 2020, marked a significant advancement in the YOLO series, emphasizing speed and accuracy. Built on PyTorch, YOLO v5 features various model sizes (s, m, l, x) to cater to different performance needs, allowing users to balance computational efficiency and detection precision. The model introduced various innovative techniques, including mosaic data augmentation and self-adversarial training, which enhanced its ability to generalize from diverse datasets.

Released by Meituan in June 2022, YOLO v6 was designed to focus on production environments and real-world application performance. This version introduces an efficient backbone and neck network alongside a more effective detection head, enhancing the model's speed and accuracy. YOLO v7, released in July 2022, aimed to set new benchmarks in real-time object detection. Key innovations include extended efficient layer aggregation networks and dynamic label assignment, improving detection speed and accuracy.

Ultralytics released YOLO v8 in January 2023, following the successful launch of YOLO v5. This latest version introduces improvements in model architecture and training approaches, enhancing detection accuracy and speeding up processing. YOLO v8's seamless compatibility with prevalent machine-learning tools and frameworks highlights its versatility and user-friendly nature [5]. YOLO v8 incorporates a more optimized backbone network, allowing better feature extraction from images. YOLO v8 uses an anchor-free detection mechanism, enabling it to receive input from objects of different shapes and sizes.

While released as of June 2024, a hypothetical YOLO v9 is anticipated to continue the trend of advancements and further refinements in model architecture to enhance speed and accuracy. It could incorporate transformer-based architectures for better handling

of varied tasks. YOLO v9 might also introduce improved techniques for multi-task learning and more sophisticated data augmentation methods [4].

The developed YOLO models can be used to detect and classify objects, especially medical waste, since they are critical. This YOLO model should overcome the challenges of misplaced and mixed waste, which are key to the problems. Thus, classification and automated sorting can help minimize pollution from medical waste and sustain community health.

We consider integrating YOLO with IoT components to develop a complete sorting machine. The YOLO model is a Deep Learning model capable of detecting specific objects and categorizing them based on prior training. This training involves using a dataset of images containing medical waste. The IoT device will process the input and convert it into signals, enabling system control through an actuator or display of information on a local or cloud platform. Our approach involves capturing input from a camera using YOLO-based detection and employing processed IoT signals to control the sorting machine.

This work integrates the YOLO model, IoT elements, and an automated sorting machine to achieve medical waste sorting and overcome the current problems. Below are this paper's contributions to sustainable health and the environment. Figure 2 illustrates this paper's contents through an organized overview.

- Enhanced accuracy in classification, compared to manual sorting.
- Improved safety for medical and waste management workers.
- Improved waste segregation for recycling.
- Intelligent decision-making with AI and IoT.
- Operational cost reduction and human labor dependencies.
- Environmental impact reduction.



Fig. 2 Research organization

- Real-time monitoring and data logging for further analysis.
- Integration with intelligent sorting machines.
- Adaptive learning and update through YOLO development.

Research review

In this section, we examine prior research on the application of Artificial Intelligence in waste image classification. Figure 3 illustrates previous studies grouped according to the type of waste. Most research focused on detecting and categorizing general waste items like metal, plastic, paper, and glass. Classification aims to recycle general waste into new products, as most general waste is recyclable, except food and biodegradable waste. It is crucial to appropriately sort electronic waste and batteries due to the presence of toxic and hazardous materials. Mixing these materials with general waste can pose environmental harm [6].

A work utilized hyperspectral imaging with an Artificial Neural Network (ANN) to detect electrical and electronic plastic. The study concentrated on e-waste, necessitating chemical separation and treatment [7]. Classifying images in Deep Learning involves convolution layers and filters known as Convolutional Neural Networks (CNN). CNN can perceive and process data to classify images through the Convolutional Layer, Activation Layer, Pooling Layer, and Fully Connected Layer. Previous research has utilized CNN to detect hazardous waste and batteries [8]. Also, CNN was used in general waste detection, classifying paper, plastic, metal, and food waste [9–11].

Also, with ANN, the researchers proposed an automated sorting system to detect and classify three types of waste, including metal, cardboard, and trash [12]. However, this research did not develop the proposed physical sorting prototype. Only the network performance was evaluated. Additionally, ANN is computationally expensive to classify compared to newer models, such as YOLO.

Researchers used a few machine learning models, such as SVM [13, 14], Decision Tree [15], Random Forest [16, 17], and VGG [18], to classify images into general waste

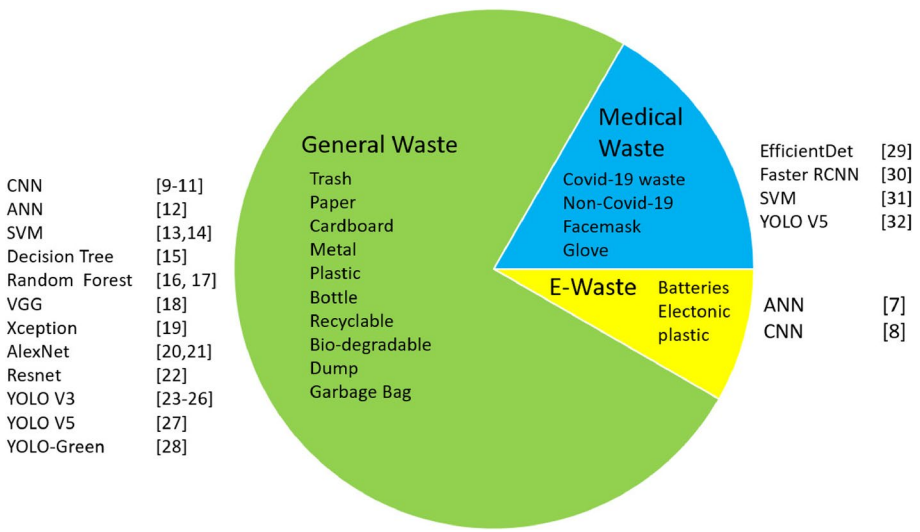


Fig. 3 AI-based methods overview

classes. The earlier generation of image classification often involved using these models, which extract features from the pictures. This work only focused on network evaluation based on general and household waste, with no mechanical integration to sorting work.

In the same scope, a work proposed a new two-layer intelligent decision system, a crossover with other models such as GoogleNet, ResNet50, Inception, and Xception. This method focused on detecting waste such as cardboard, glass, metal, and trash [19]. This network fusion and benchmarking system can be applied to the expanded version of the waste application. Compared to our work, we focused on detecting and sorting medical waste.

Due to the network's ability to automatically learn features, Deep Learning models perform better on complex tasks such as image classification. In previous works, the researchers used AlexNet [20, 21], ResNet [22], and YOLO models, focusing on the same scope: classifying general waste. YOLO v3 [23–26], YOLO v5 [27], and YOLO-Green [28] are the versions of YOLO models that were used in the previous image learning and object detection.

YOLO is the newest model for real-time object detection, providing the adaptability for camera-based detection and detecting objects in each video frame. It is easier to utilize the YOLO model since it is adaptive and updated in the network to improve accuracy and speed. It can also potentially be used with a real-time camera. In addition, it can be implemented in a microcontroller for further real-time testing and development. Previous YOLO research discusses its capability in detection, but not in actual sorting applications.

Focusing on the scope of medical waste, a few studies have used different approaches to detect COVID-19 and non-COVID-19 waste [29–32]. The COVID-19 waste is limited to facemasks and gloves. The application was discussed, but the work was not implemented according to the trained methods. These methods could be utilized to detect waste in real-time with the help of robots and other systems to minimize pollution. However, no developed sorting system was presented to manage absolute waste.

The emergence of medical waste is a fresh concern following the COVID-19 pandemic. It is leading to new sources of pollution in the environment, impacting both land and the ocean. Even though we are now in a post-pandemic scenario, managing medical waste remains challenging due to manual segregation. The autonomous waste sorting process is advantageous for the current situation and will also be beneficial in future pandemics.

This work will leverage the elements, including the YOLO algorithm, IoT system, and mechanical parts, in developing the prototype. The process includes algorithm training, testing, motors, mechanical development, and programming. Based on the previous work reviews, the contributions and gaps are below.

Previous contribution.

- Enhancing the detection and classification by proposing a new network.
- Improving accuracy and speed through the training and testing of the network.
- Focused on general and household waste such as paper, metal, glass, plastic, recyclable, biodegradable, and e-waste.

Previous works' gaps.

- Limited discussion on actual implementation and prototype development.
- The trained Deep Learning models were not implemented on the sorting machine, so medical waste sorting was not assessed.
- There is no complete system that integrates an AI model, mechanical parts, and an IoT system for further benefits.
- The datasets only covered general waste objects.
- The precision and accuracy of the models from previous work are low, requiring YOLO model evaluation, specifically using a medical waste dataset.

Our work contribution:

- Integrating the YOLO model, mechanical sorting mechanism, and IoT into a complete system is especially crucial for medical waste management.
- We are separating the class into facemasks, gloves, syringes, and sharp waste for recycling and to avoid accidents or injuries from sharp waste.

Methodology

YOLO neural network

Each YOLO version has its specialty, based on the blocks and modules introduced in the algorithm. Convolutional layers are the primary filters that extract and filter the features in an image. The images undergo each layer in the YOLO model to learn and generalize the objects based on notable features such as shape, color, and others.

The YOLO model was structured into three sections: backbone, neck, and head. The backbone consists of blocks and modules to extract features and filter the unimportant information. Once the feature data was extracted, it was brought to the neck for further feature fusion, and multiscale data split for detection. The head section consists of a detection block to perform detection and classification. Table 1 outlines the key layers and modules introduced in each YOLO model.

Table 1 YOLO algorithm parameters

YOLO version	Key layers/modules	Function
YOLO v5	Conv C3 SPPF	First baseline PyTorch YOLO C3 is for feature extraction SPPF is for multiscale detection
YOLO v6	Conv SPPF	Mainly used Conv layers for feature extraction
YOLO v7	Conv RepConv	Re-parameterizable convolution is used to increase speed while maintaining accuracy
YOLO v8	Conv C2f SPPF Concat	Improved block with lightweight C2f. C2f replaces C3 for improved feature extraction
YOLO v9	Conv RepNCSPELAN4 SPPELAN ADown	Adown highlights the feature with great attention. SPPELAN and RepNCSPELAN4 improve the feature fusion, increasing accuracy

YOLO training

Five YOLO versions (v5, v6, v7, v8, and v9) were used for the training, with 6760 images as a dataset. The dataset featured various classes: facemasks, syringes, gloves, surgical blades, needles, vials, and hands. The datasets for sharp object detection, including blades, needles, and vials, needed to be detected similarly to masks, syringes, and gloves.

The training was conducted using Google Colab's online notebook, which facilitated the integration of key components such as the prepared dataset in Roboflow, the model from the repository, and the online runtime accelerator. The YOLO algorithm process commenced with dataset preparation, image labeling, training, and mAP evaluation.

Roboflow proved to be a valuable online tool for labeling and storing images. The labeled images were exported as a dataset and used for training in the Colab Notebook. Additionally, the dataset can be updated by adding more images and expanding the classes. For future improvements, the training can be repeated using the same dataset. Figure 4 depicts the labeling process in Roboflow, while Fig. 5 highlights an overview of the instances in the dataset for training, validation, and testing.

The images are varied based on the background and the object's condition. The objects in images can be in a simple blank background or a filled background, such as the waste on the road, sand, and stacked objects. The waste can also be in regular shape and condition or folded, crumpled, and cut. From this variation, the YOLO model can learn how to generalize objects and classify objects from different views. The dataset contained such objects, but they were mixed and not counted per variation.

The YOLO training algorithms were configured with specific properties, such as batch size, number of epochs, and augmentation techniques. A detailed overview of the YOLO training process can be observed in Table 2. The batch size refers to the number of iterations used for model updating, and in our case, 6760 images were divided into 16

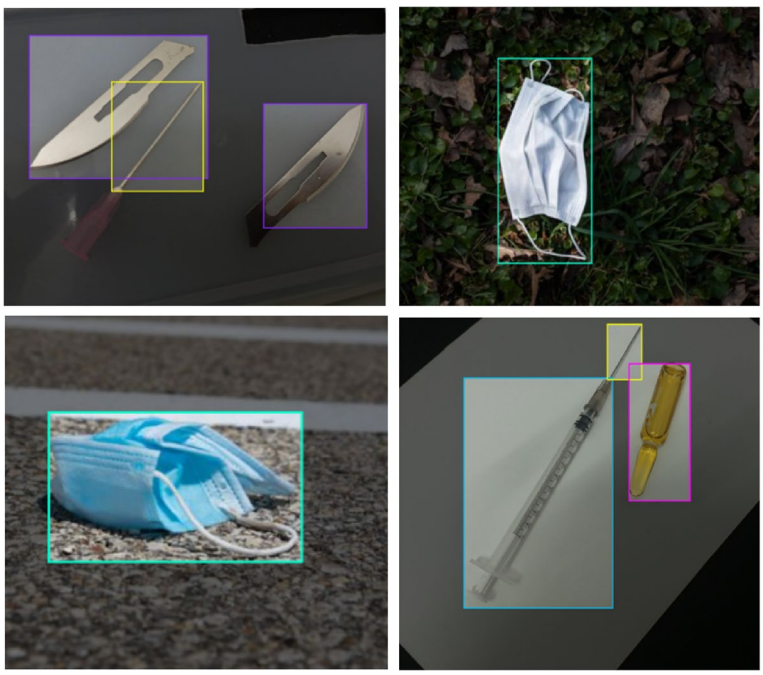


Fig. 4 Annotating images in Roboflow



Fig. 5 Dataset overview in Roboflow

Table 2 YOLO configurations

YOLO model	V5	V6	V7	V8	V9
Batch	16	16	16	16	16
Epoch	25	25	25	25	25
Image size	416	800	640	800	800
Learning rate	0.01	0.01	0.01	0.01	0.01
Optimizer	SGD	Auto	Group	Auto	Auto
Augmentation	Mosaic Flip	Mosaic Flip	Mosaic Flip Mix up	Mosaic Flip	Mosaic Flip Mix up
Layers	153	195	415	225	621
Parameters	2,509,829	4,238,837	37,228,920	11,138,309	25,442,469
Gradients	2,509,813	4,238,821	37,228,920	11,138,293	25,442,453
GFLOPs	7.2	11.9	105.2	28.7	103.2

batches. Each batch triggers a weight update in the model. Epochs denote the number of complete training runs, with all 6760 images being trained 25 times.

The augmentation was employed to enhance the diversity of the training data, utilizing techniques such as shear, scale, flip, and mosaic. The augmentations were not applied during dataset processing in the Roboflow platform, but the augmentations were already set in the YOLO models. Thus, they were used during the training process.

Data augmentation techniques like shear, scale, flip, and mosaic create diverse training instances, decreasing overfitting and the model's dependence on specific features. Overfitting occurs when the model struggles to generalize object characteristics due to limited input. Through data augmentation, the model becomes better at generalizing to unseen data, enhancing accuracy and precision.

YOLO evaluation metric

The YOLO algorithm is used for real-time object detection. It works by dividing the image into a grid and then predicting the bounding boxes and class probabilities within each grid cell. The mean average precision (mAP) is a comprehensive evaluation metric for YOLO models. This is due to the fact that the YOLO model can detect multiple

objects within an image. The precision and recall of the predicted bounding boxes reflect the model's performance in locating specific targets and classifying samples. The average precision provides an overall assessment of the performance of each target or class. These evaluation metrics are calculated using the following formulas:

$$\text{Precision}, = \frac{\text{TP}}{\text{TP} + \text{FP}}, \tag{1}$$

$$\text{Recall, R} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{2}$$

$$\text{Average Precision, AP} = \int_0^1 P.\text{RdR}, \tag{3}$$

$$\text{Mean Average Precision, mAP} = \frac{1}{N}\sum_{i=1}^{N} AP_i. \tag{4}$$

The metric mAP is utilized as a standard benchmark in object detection tasks, facilitating the comparison of YOLO model performance. mAP provides a well-rounded assessment of the model's capability to identify and categorize objects, such as medical waste, in our specific case. A singular accuracy metric may not adequately reflect the model's performance across all waste categories. By averaging AP values for all classes, mAP ensures a comprehensive evaluation of the model.

Electronic hardware

Figure 6 illustrates the prototype's IoT connection. The Raspberry Pi 4 is linked to the camera to capture images for object detection using the YOLO v8 model. When the system identifies the object class, the machine uses motors and a sorting mechanism to place them in categorized bins. If multiple objects are detected in the same image, the sorting signal is not transmitted to the motor controllers, resulting in individual waste sorting.

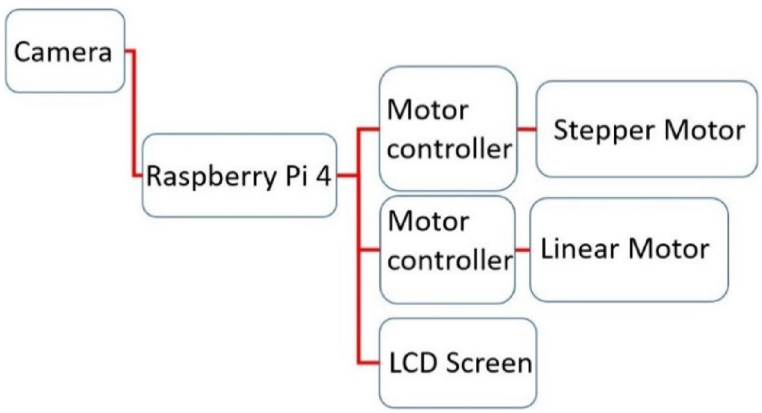


Fig. 6 Electronic parts connection for the prototype

Using YOLO v8 with a Raspberry Pi is a cost-effective approach for developing the prototype, enabling the detection, classification, and sorting of medical waste. Such equipment deployment represents the initial phase of development, facilitating the testing of sorting functionality and paving the way for determining challenges and opportunities for enhancement in actual and large-scale implementation. Additionally, YOLO v8 is highly resource-efficient, allowing it to run smoothly on Raspberry Pi hardware. Figure 7 illustrates the electronic hardware during development, including the Raspberry Pi and the motor drivers.

Prototype design

Figure 8 illustrates the key mechanical components of this prototype. The linear motor and stepper motor performed the linear and tilting motions, with the connector and carrier being crucial components to link both motors. Figure 9 depicts the direction of the plate motion, both linear and tilted. The carriage followed the line of the aluminum profile below, moving linearly and being driven by the linear actuator.

Two motors moved the waste-containing plate to the designated bin location. In the prototype, three bins were set up, and a linear motor moved the plate in a straight line, tilting it left or right to deposit waste into different bins. A stepper motor tilted the plate at a 35° rotation before returning it to its original position for further sorting.

Programming work

Figure 10 illustrates the flowchart of the program that was used in the Raspberry Pi controller to detect the object through the camera and drop it into the classified bin. Python was used in Geany coding software and the Raspberry Pi environment. The trained YOLO v8 model produced a file to be used in programming work: the weight file. The weight file contains the parameters trained by the YOLO training.

The program was initiated by detecting images from the camera and then utilizing the YOLO model for classification. YOLO analyzed the image, generating output that included labels and bounding boxes. Based on the YOLO object detection results, loops and if-else statements were employed to control the motor driver. Waste items were directed to different bins based on whether they were identified as facemasks, gloves, or syringes. This process was repeated for each image frame and ceased when the code's

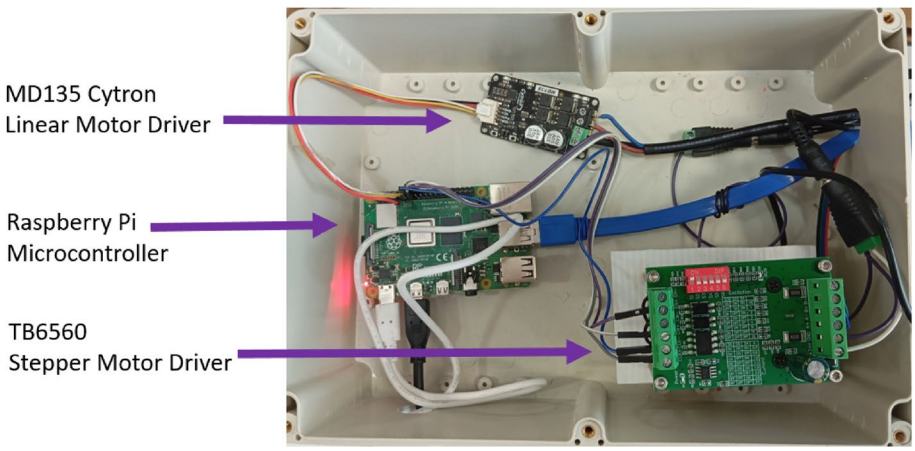


Fig. 7 Electronic hardware

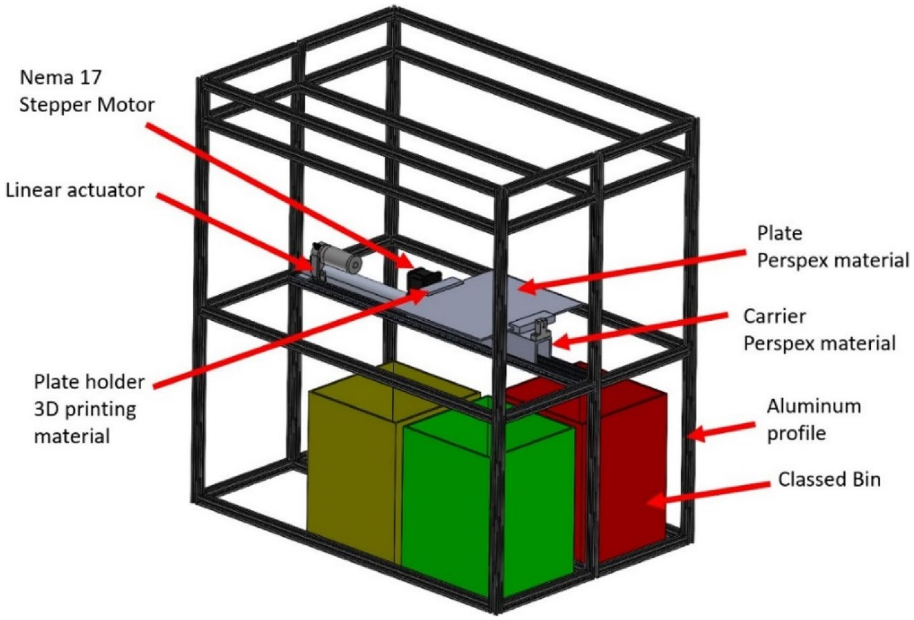


Fig. 8 Mechanical parts of the prototype

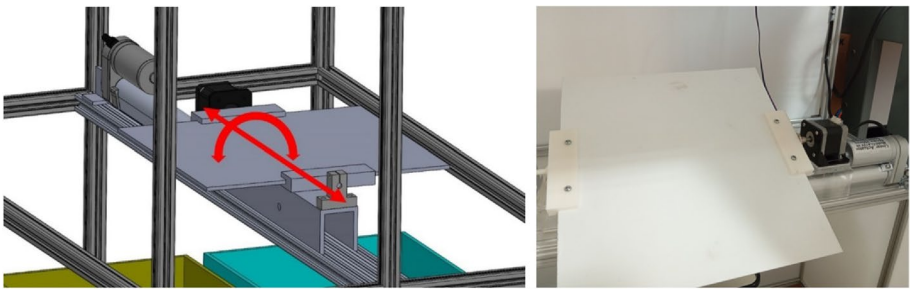


Fig. 9 Plate design with mechanism direction (drawing and actual)

terminal was closed. The loop was necessary to ensure continuous waste sorting after the user disposed of waste into the machine. The if-else statements were implemented to determine the appropriate classes for waste sorting in the machine.

IoT development.

The IoT element was incorporated into the program to publish the sorting information to the IoT cloud for monitoring purposes. Data such as current detection output, the waste counts for each class, and the bin capacity are essential to monitor. However, based on the additional sensors, the data can be expanded into other parameters in future work.

The Blynk platform was used to send the data and build the dashboard. A Python library named requests is the key to publishing the data, including integers and strings. In the system, we count the detected waste per class and estimate the bin capacity based on the count. The classes are facemasks, gloves, syringes, and sharp waste. The sharp waste consists of needles, blades, and other items. This item will be placed in the same bin named Sharp Waste Bin. Then, after the sorting, the count and capacity were

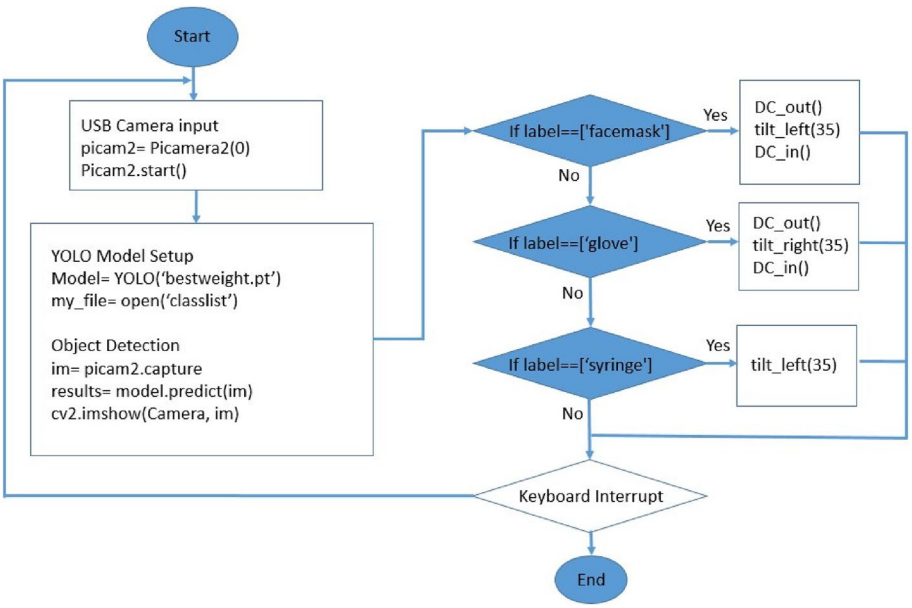


Fig. 10 Flowchart of the program

```
# Send count and capacity to Blynk
try:
    requests.get(f"{BLYNK_URL}&{VPIN_COUNTS[class_name]}={waste_counts[class_name]}")
    requests.get(f"{BLYNK_URL}&{VPIN_BINS[class_name]}={bin_capacity[class_name]}")
except:
    print("❌ Failed to update Blynk for", class_name)
```

Fig. 11 The request.get command for data publishing

updated and published to the Blynk cloud through a "request.get" command as in Fig. 11 below. The transmission protocol is HTTP, since the requests library was used. This data publication used Blynk TOKEN as an authentication setup to ensure the data was safely sent in the right direction.

Sorting accuracy testing

Once the utilization of the YOLO model and the development of the prototype were complete, the testing was conducted to assess the accuracy of the sorting. We prepared 100 waste samples from each class to be placed on the plate for camera detection. After that, the samples will be classified and moved to the bins based on their category, through the Raspberry Pi command and the motors function. Sharp waste, including blades, needles, and vials, will be placed together. Others, like the glove, syringe, and facemask, will be placed in respective bins. The lighting and the camera quality were the constant variables, where the 4 K camera and LED light with 250 lx intensity were used in the prototype.

The samples vary between a regular mask, a crumpled mask, a normal glove, a crumpled glove, a small syringe, a normal-sized syringe, a broken vial, a filled vial, a needle, and a blade. The shape or condition of the waste is varied to observe the different accuracy between them, and to see the effectiveness of the YOLO in generalizing the waste.

Figure 12 below displays an example of the samples. Data like true positive, false positive, and false negative were also collected for the accuracy calculation.

Results and discussion

This section discusses the results, including YOLO model training, actual detection and sorting testing, final prototype specification, IoT dashboard, and energy consumption estimation.

YOLO training results

Table 3 below presents the AP for different classes and the mAP at 0.5 Intersection over Union (IoU) for various versions of the YOLO model. The performance metrics include AP for blade, facemask, glove, hand, needle, syringe, and vial.

YOLO v8 and YOLO v9 exhibit the highest precision, recall, and mAP performance, with YOLO v8 slightly outperforming in training time compared to YOLO v9. YOLO v5 has the shortest training time but lower performance metrics, while YOLO v7, despite good performance, has the longest training time. YOLO v5 and v6 present lower AP and mAP values, particularly for the Needle and Syringe classes.

While YOLO v8 and YOLO v9 perform exceptionally well across most classes, YOLO v8 achieves a slightly higher mAP at 0.5 IoU, 0.98. There are a few reasons why YOLO v8 might outperform YOLO v9. YOLO v8 may have benefited from more optimal hyperparameter settings for this particular dataset. The architecture of YOLO v8 might align better with the dataset characteristics, leading to better performance. The changes introduced in YOLO v9 did not translate to performance gains for this dataset. YOLO v9 might be more prone to overfitting due to its complexity, leading to slightly lower generalization performance than YOLO v8.

The comparison table from Table 4 provides an overview of various YOLO models and their performance metrics, specifically mAP, the number of classes they can detect, and the size of the datasets, all focusing on detecting medical waste or surgical objects. From our work, YOLO v8 and YOLO v9 demonstrate high mAP values of 0.98 and 0.978,

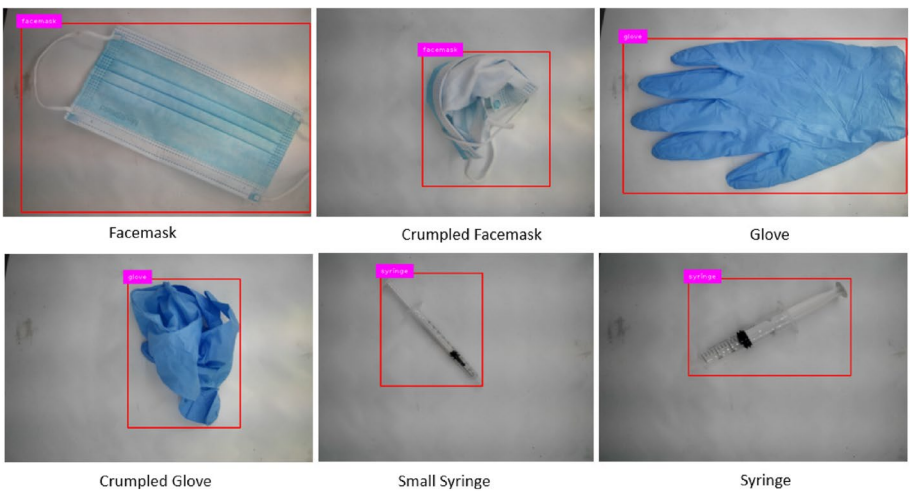


Fig. 12 Samples for accuracy testing

Table 3 YOLO training results

YOLO model		V5	V6	V7	V8	V9
Average precision AP	Blade	0.974	0.975	0.986	0.995	0.994
	Mask	0.934	0.96	0.957	0.981	0.982
	Glove	0.914	0.966	0.976	0.979	0.984
	Hand	0.94	0.954	0.948	0.988	0.988
	Needle	0.877	0.894	0.904	0.959	0.95
	Syringe	0.829	0.871	0.894	0.97	0.968
	Vial	0.937	0.927	0.942	0.99	0.982
mAP @ 0.5 IOU		0.915	0.935	0.944	0.98	0.978
Precision		0.93	0.909	0.917	0.958	0.961
Recall		0.861	0.887	0.898	0.963	0.965
Training time (h)		2.518	4.232	18.522	4.431	8.726
F-1 score		0.89	0.9	0.91	0.961	0.96
Confidence		0.436	0.385	0.399	0.491	0.45
FPS		52.269	56.36	55.867	52.692	26.867

Table 4 Results comparison

Year	YOLO model	mAP	Medical waste classes	Datasets
2024 Our work	YOLO v8	0.98	7	6760
	YOLO v9	0.978		
2024	YOLO v8	0.824	3	1133
2022	ME-YOLO (modified v5)	0.972	5	2500
	YOLO v5	0.961		
2022	YOLO-X	0.925	3	1153
	YOLO v3	0.854		
	YOLO v5	0.848		

respectively, handling seven classes with a dataset of 6760 images. Another instance of YOLO v8, evaluated on a smaller dataset of 1153 images with three classes, suggests a lower mAP of 0.824 [33].

ME-YOLO, a modified version of YOLO v5 from a research paper, achieves an mAP of 0.972, detecting five classes over a dataset of 2500 images [34]. YOLO-X achieves an mAP of 0.925 on three classes with 1153 images [35]. This comparison highlights the varying performance of YOLO models depending on their configurations and the datasets used for evaluation, all within the context of detecting medical waste and surgical objects.

Besides the common medical waste (facemasks, syringes, and gloves), our work also focused on sharp waste, including surgical blades, vials, and needles, which were not addressed in other papers. Conversely, the other papers included classes not covered in our work, such as goggles, biohazard waste, PPE suits, and face shields.

Figure 13 highlights the confusion matrix of YOLO v8. The obtained confusion matrix for YOLO v8 is a normalized representation of the model's performance across different classes. Each cell in the matrix indicates the proportion of predictions for a given class (rows) that were correctly or incorrectly classified as another class

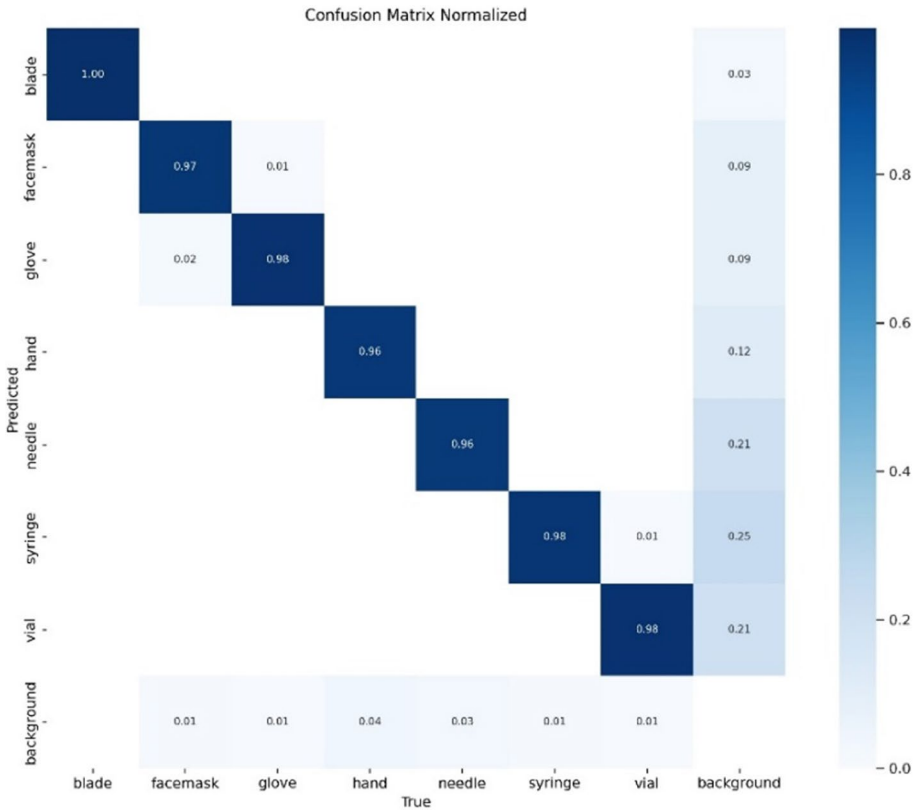


Fig. 13 Confusion matrix of YOLO v8

(columns). The Blade class has a perfect classification with a positive rate of 1.00. For other courses, the true positive rate (TP) ranges between 0.96 and 0.98, with a misclassification rate between 0.01 and 0.04. Overall, the confusion matrix indicates that YOLO v8 performs exceptionally well with high accuracy for most classes.

Figure 14 illustrates the P-R curve, highlighting the AP of all classes derived from precision and recall. The curve also displays the mAP for all classes and the AP of each class. These curves, built from the training data, reflect the AP of each class through the area under the P-R curve.YOLO v8 obtained significant results with 98% mAP, 0.958 precision, and 0.963 recall. These high metrics indicate that the sorting machine should accurately identify and sort waste, reducing misclassifications and minimizing the need for manual checking and re-sorting. This model was selected for implementation in medical waste sorting work.

The trained model was then used for validation, as shown in the collected images in Fig. 15. The trained YOLO model successfully detected the object, with the labeled class. The objects were detected and labeled with the confidence score, which reflects how effectively the YOLO model recognizes medical waste. Most of the detections achieve a high confidence score, with minimum misclassification.

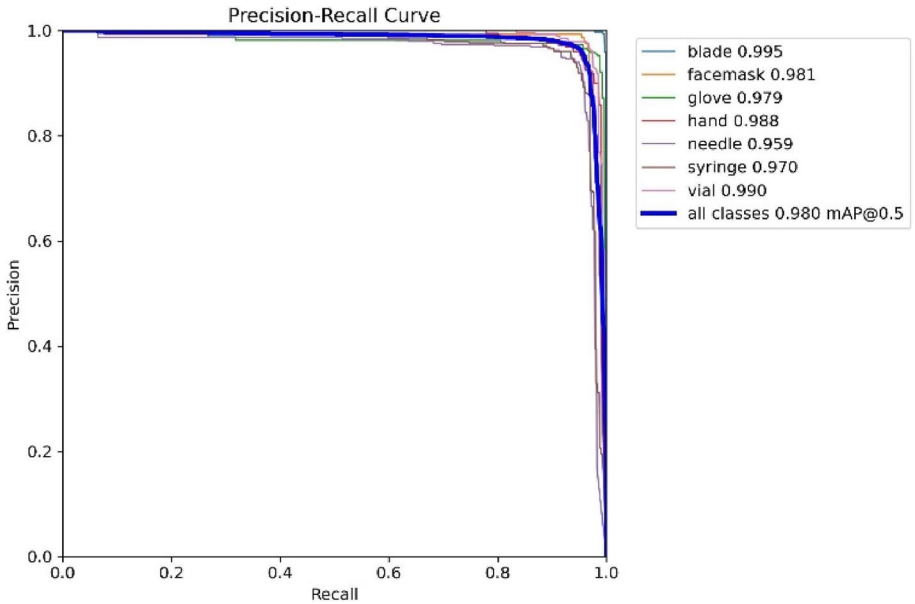


Fig. 14 Precision-recall curve of YOLO v8



Fig. 15 Image validation of YOLO v8 training

Accuracy test with YOLO v8

The YOLO v8 model was used in programming work to detect and classify the waste based on the camera. Upon the classification output, the waste is then moved to the designated bin based on class. From Fig. 16, the camera is utilized to output the detection of medical waste. The system can identify and categorize waste into the appropriate class using the code and other necessary YOLO components.

One hundred waste samples from each class were used for the sorting accuracy testing. The waste was placed on the plate under the camera and identified by the YOLO v8 model. Then, it was sorted into the bin based on the label output. The classified bins are for facemasks, syringes, gloves, and sharp waste.

Table 5 shows the accuracy results for overall testing and each waste type. Glove, syringe, and vial have lower detection accuracy and require feature extraction, dataset expansion, and YOLO algorithm improvement. A facemask has high accuracy, while crumpled mask detection has lower accuracy since it has features similar to a glove when crumpled and folded, mimicking the same shape and color. Thus, further study is required to help the model generalize their differences. The vial has a small

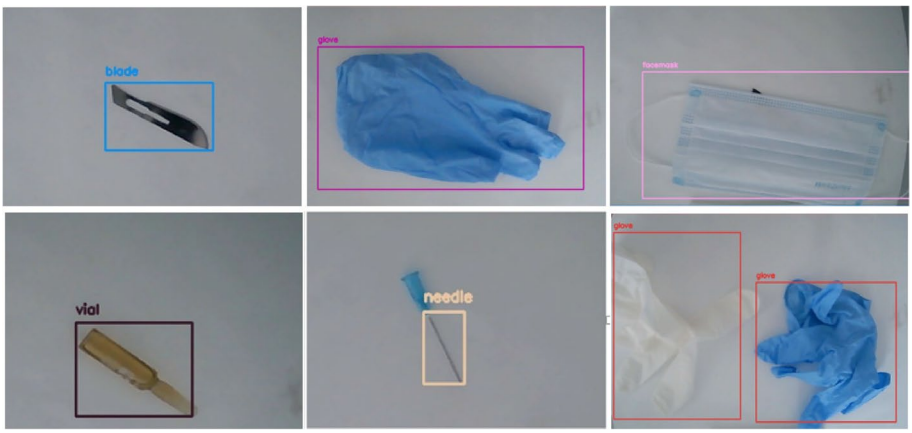


Fig. 16 Object detection by using the camera

Table 5 Detection and sorting accuracy

Class	TP	FP	FN	Accuracy (%)
Facemask	98	1	1	98
Crumpled masks	93	6	1	93
Glove	94	5	1	94
Syringe	94	5	1	94
Needle	98	2	0	98
Blade	97	3	0	97
Filled vial	91	7	2	91
Empty vial	88	7	5	88
		Average detection and sorting accuracy		93.75

size and a reflecting surface, making it challenging for the model to recognize. This needs attention, and the model needs some customization to adapt to this feature. The needle and blade have good detection accuracy since they have a rigid shape and features.

Final developed prototype

The successful development of the prototype involved integrating mechanical, electronic, and software components. A camera was positioned atop the sorting machine to capture images of the waste-filled plate. Users could place objects on the plate through the inlet. The electrical box contains the devices and wires connecting the Raspberry Pi, motor drivers, LCD screen, lightning device, and camera. Figure 17 depicts the complete sorting prototype. Figure 18 illustrates the prototype body's internal components, showing where the inlet, plate, bins, motors, and camera are. The plate will be activated when a waste class is declared, leading to the waste being deposited into the appropriate bin. Table 6 summarizes the prototype configuration, including size, speed, energy consumption estimation, and other specifications.



Fig. 17 Sorting prototype

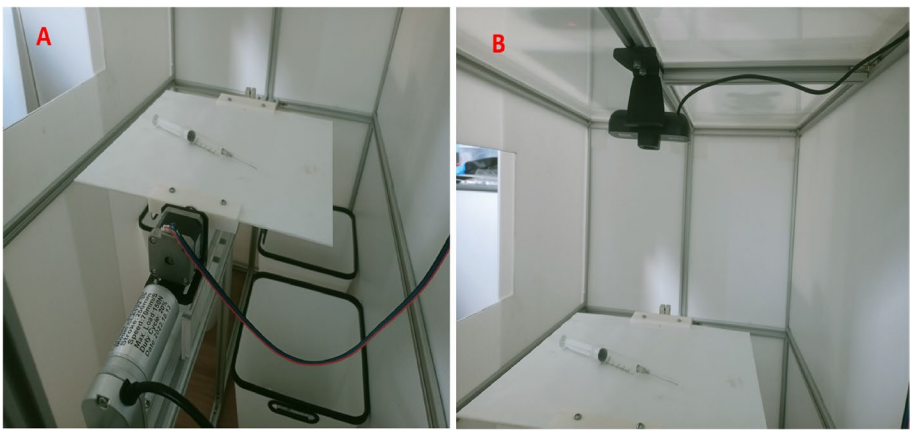


Fig. 18 The prototype components

IoT monitoring system

The IoT system was added to the sorting prototype to publish the informatics data to the dashboard for monitoring. The Raspberry Pi runs the code to send the data to the Blynk hub. Figure 19 depicts the developed dashboard in Blynk, where the user can monitor the bin capacity, the waste count, and the current detection.

For future work, other valuable data can be added, including the real-time camera output and other input from useful sensors for medical purposes. The liquid and metal

Table 6 Prototype summary

Details of the prototype	Results
Size	120 × 90 cm
Plate size	22 × 20 cm
Camera type	4 k USB camera
Lightning	247 to 250 lx
Sorting bin number	4
Detection time	10 to 11 s
Sorting time	20 s
Complete cycle time	31 s
Sorting accuracy	93.75%
Power consumption estimation	55 W

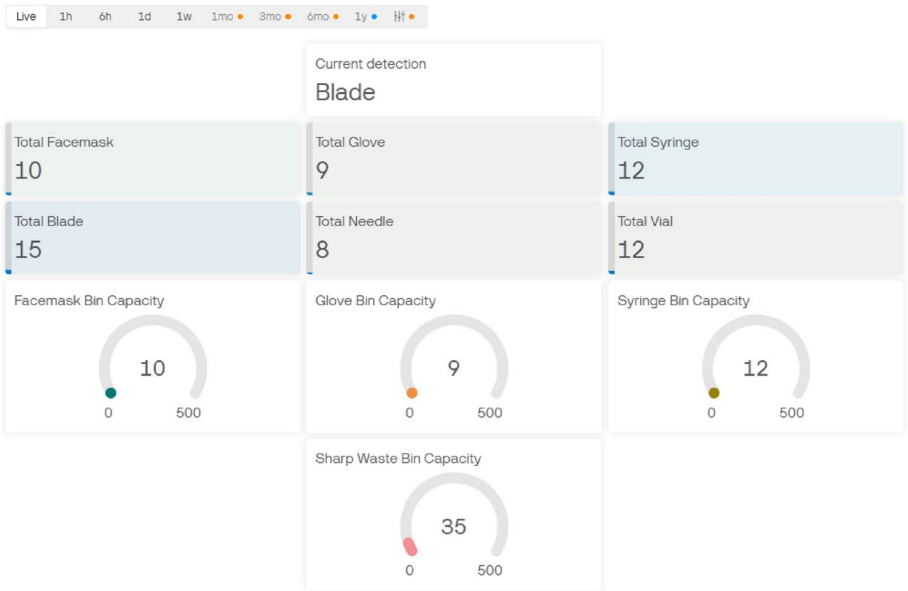


Fig. 19 IoT dashboard

detectors can also be used and studied to detect metal waste and recognize fluids such as blood to monitor the contamination level inside the sorting machine. UV sterilization should be added to control and minimize the contamination risk when collecting the waste.

Prototype energy estimation

Table 7 is a breakdown of the energy estimation regarding each component. The power was estimated based on its voltage and current input. Total energy consumption for an hour is 55 Wh. Assuming eight hours of daily operation, the daily energy is 440 Wh. Using renewable energy as an alternative is possible, minimizing the primary energy consumption and saving costs. The solar energy assessment was set, and the battery capacity was calculated based on the weather in Malaysia. Table 8 shows this sorting

Table 7 Energy estimation per component

Component	Power (W)
Raspberry Pi 4	6
Camera	1.05
LCD Screen	3.5
NEMA 17 stepper	14.4
12 V linear actuator	18
2 motor controllers	3
Total	~ 46
Added 20% margin	~ 55 W

Table 8 Solar energy consumption for future work

Component	Calculated and recommended specification
Solar panel	120–150 W
Battery	12 V, 50 Ah (LiFePo)
Charge controller	MPPT, 12 V, 15–20 A
Buck converter	12 V—5 V, 3 A min (for Raspberry Pi)

machine prototype's solar system setup plan for future implementation in medical waste management.

Further consideration

This work only focused on integrating the YOLO model with other essential elements, including the Raspberry Pi and the sorting mechanism. Hence, a complete system was developed to sort medical waste into classified bins. This work will be further improved in many aspects. Below are the limitations and considerations in implementing the automated sorting system in medical waste management.

YOLO training improvements

Other YOLO improvements are required, including pre-processing, augmentations, and other modifications. Data augmentation techniques generate various training examples, helping prevent overfitting by reducing the model's reliance on specific patterns or features in the original training data. Various augmentations can be utilized to observe which augmentation technique provides the highest accuracy. Architecture customization, including layer modification, can also improve feature extraction. The increasing number of images with class balance also allows the model to extract more features, improving classification.

Alternative sensors and cameras

During the sorting process, a few factors can significantly impact the accuracy of the object prediction. The plate's waste size, shape, and placement variations can cause the camera to collect the images and classify the waste. Smaller objects like needles and

broken vials may be complex for the model to recognize. Conversely, larger objects might partially obstruct the camera's view, causing incomplete detections. The camera might not capture a clear image if the waste is poorly positioned. For improvement, multiple cameras can be added at different angles to capture the medical waste from various perspectives while handling the small or large objects on the plate. Conditions like overlapping, folded, and tilted waste can also be managed with better recognition.

Sorting medical waste based on camera detection only produces one classification output. We also require additional input to support the classification, such as metal and liquid detection, to detect the waste differently. Additional sensors may be beneficial for further study, enhancing the sorting performance and monitoring.

Actual sorting implementation

From the actual implementation perspective, the medical waste sorting machine can be used in more extensive medical facilities. At the same time, a few elements need to be evaluated and considered to achieve efficient management. The quality of the camera for image capture needs to be improved to get better object detection. The controlling system, power consumption analysis, and mechanical setup need to be updated to ensure suitable types of equipment for the actual application. This research sorting prototype is the first step of integrating YOLO and a sorting machine, while further assessment is required for a more extensive application.

Different sorting mechanisms can be used to implement recycling facilities. Delta robots with conveyor belts benefit large-scale sorting, such as in recycling factories, where we can sort multiple types of waste on time. Unsorted waste can be received from waste collection. However, it requires proper development and a high cost of equipment. Further study can be conducted on the Delta robot analysis (motion simulation and gripping solution) for medical waste sorting applications.

The sorting machine must be reliable in overseeing substantial amounts of waste. The main controllers and motor need to operate for a long time. The system requires a high-precision microcomputer, a GPU to process the task, and high-quality sensors and cameras for industry-level operation. Safety features, authority recognition, management guidelines, and training are the key considerations for implementing the system and ensuring its safe operation.

Contamination control

After sorting, the waste from the bins will be collected for further processing, such as treatment and transportation. We must ensure the bin collection is protected from infections and accidents. UV sterilization must be added to the sorting machine to minimize this risk and ensure safe bin collection. In addition, the pickup robot can help humans perform this task.

Research limitations

This research found a few limitations. The sorting accuracy is high but still requires improvement. There are still misclassifications or detection errors, which need to be minimized before they can be implemented in actual operation. The above considerations, such as multiple cameras and sensors for high detection confidence and avoiding

mixed waste, help minimize the error. Thus, human interventions can be reduced. The reliability must also be improved by further suitable selection of micro-computer, controller, and motors. The critical element is safety, where contamination controls must be considered. Finally, this sorting work only sorts one waste at a time, requiring time and many sorting cycles to complete the task. Other mechanisms can be helpful in multiple waste sorting and management.

Conclusion

Theoretically, this research integrated AI, IoT systems, and sorting mechanisms in medical waste management. YOLO object detection was introduced and utilized in the developed sorting machine, with a real-time monitoring system through IoT and the Blynk platform. This merged system offered highly accurate autonomous medical waste sorting through computing solutions, solving manual sorting problems.

This work developed an automated sorting prototype for medical waste management and achieved 93.75% sorting accuracy. It proves that healthcare can minimize manual handling of contaminated, hazardous, and sharp medical waste. With a monitoring system, waste management becomes more auditable, transparent, and reachable. Efficient and safe waste disposal is the implication, especially during the post-pandemic, sustaining community health and saving the environment.

This work contributes to object detection, specifically for medical waste, which is unique in shape, feature, and texture compared to the general waste. The IoT integration enables live monitoring, helping in further analysis and evaluation among medical staff in handling medical waste. Not only does the system conduct detection as in previous works, but it also performs sorting using a Raspberry Pi. The system is designed with cost-effective components, making it implementable in healthcare.

The developed system has several advantages, including reducing contaminated waste exposure, avoiding misclassification from manual sorting, and an IoT dashboard for data logging, analysis, and auditing. This can be implemented in small clinics and large healthcare centers, requiring further consideration.

Although the utilized YOLO model is highly accurate, it still has a detection error and class confusion. The algorithm must be improved to minimize misclassifications. Other than that, safety, cleanliness, and efficiency are the key elements to improving the prototype and solving the limitations. These elements are not yet included in the developed system, but can be added in future work.

This research can be expanded by studying alternative sensors to detect medical waste and obtaining new informative data from different perspectives. This can add value to research, not relying on just one method. In addition, a new sorting mechanism can be implemented and assessed on the current system to compare the effectiveness of linear sorting with others, such as delta robots and conveyor belts.

Abbreviations

IoT	Internet of Things
AI	Artificial intelligence
YOLO	You Only Look Once
AP	Average precision
ANN	Artificial neural network
CNN	Convolutional neural network

SVM Support vector machine
VGG Visual Geometry Group
LCD Liquid crystal display
TP True positive
FP False positive
FN False negative

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Authors' contributions
MHM prepared and executed the methods, including dataset collection, model training, and prototype development, and was a major contributor to writing the manuscript. HM contributed to conceptualization, supervision, and project administration. SSHH contributed to project planning, aim formulation, data reviewing, and funding acquisition. All authors read and approved the final manuscript. MZB contributed to conceptualization and method expansion.

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Data availability
The data of this research is available in the CodeOcean repository: https://codeocean.com/capsule/6755906/tree.

Declarations

Competing interests
The authors declare that they have no competing interests.

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References

1. Das A. K., Islam M. N. , Billah M. M., and Sarker A. COVID-19 pandemic and healthcare solid waste management strategy – a mini-review, Sci Total Environ., vol. 778, no. January, p. 146220, Jul. 2021, https://doi.org/10.1016/j.scitotenv.2021.146220.
2. Asim N, Badiei M, Sopian K (Aug.2021) Review of the valorization options for the proper disposal of face masks during the COVID-19 pandemic. Environ Technol Innov 23:101797. https://doi.org/10.1016/j.eti.2021.101797
3. Ur A, Saeed M, Abed M (2023) Fppsv-NHSS : Fuzzy parameterized possibility single valued neutrosophic hypersoft set to site selection for solid waste management. Appl Soft Comput 140:110273. https://doi.org/10.1016/j.asoc.2023.110273
4. Nelson J. What is YOLO? The Ultimate Guide [2024], *roboflow*, 2024. https://blog.roboflow.com/guide-to-yolo-models/ (accessed Jul. 09, 2024).
5. Jocher G., Chaurasia A., and Qiu J., YOLOv8 by Ultralytics, *GitHub*, 2023. https://github.com/ultralytics/ultralytics (accessed Jul. 01, 2024).
6. ProgressKemp, e-waste (Management and Handling) Rules, *ProgressKamp*. http://www.moef.nic.in/downloads/public-information/Modified Draft E-waste.pdf (accessed May 22, 2024).
7. Tehrani A and Karbasi H. A novel integration of hyper-spectral imaging and neural networks to process waste electrical and electronic plastics, in *2017 IEEE Conference on Technologies for Sustainability (SusTech)*, Nov. 2017, vol. 2018-Janua, pp. 1–5. https://doi.org/10.1109/SusTech.2017.8333533.
8. Hua D,, Gao J., Mayo R., Smedley A., Puranik P, and Zhan J. Segregating hazardous waste using deep neural networks in real-time video, in *2020 10th Annual Computing and Communication Workshop and Conference (CCWC)*, Jan. 2020, pp. 1016–1022. https://doi.org/10.1109/CCWC47524.2020.9031194.
9. Patil A., Tatke A., Vachhani N., Patil M., and Gulhane P. Garbage classifying application using deep learning techniques, in *2021 International Conference on Recent Trends on Electronics, Information, Communication & Technology (RTEICT)*, Aug. 2021, pp. 122–130. https://doi.org/10.1109/RTEICT52294.2021.9573599.
10. Sultana R., Adams R. D., Yan Y., Yanik P. M., and Tanaka M. L. Trash and recycled material identification using convolutional neural networks (CNN), in *2020 SoutheastCon*, Mar. 2020, vol. 2020-March, pp. 1–8. https://doi.org/10.1109/SoutheastCon44009.2020.9249739.
11. Gondal AU et al (Jul.2021) Real time multipurpose smart waste classification model for efficient recycling in smart cities using multilayer convolutional neural network and perceptron. Sensors 21(14):4916. https://doi.org/10.3390/s21144916
12. Abed M., Mahmood M., and Abdulhasan J. Automated waste-sorting and recycling classification using artificial neural network and features fusion : a digital-enabled circular economy vision for smart cities, Multimed Tools Appl., no. 0123456789, 2022, https://doi.org/10.1007/s11042-021-11537-0.
13. Ramsurrun N., Suddul G., Armoogum S., and Foogooa R. Recyclable waste classification using computer vision and deep learning, in *2021 Zooming Innovation in Consumer Technologies Conference (ZINC)*, May 2021, pp. 11–15. https://doi.org/10.1109/ZINC52049.2021.9499291.

14. Puspaningrum A. P. *et al.*. Waste classification using support vector machine with SIFT-PCA feature extraction, in *2020 4th International Conference on Informatics and Computational Sciences (ICICoS)*, Nov. 2020, pp. 1–6. https://doi.org/10.1109/ICICoS51170.2020.9298982.
15. Sami KN, Amin ZMA, Hassan R (Dec.2020) Waste management using machine learning and deep learning algorithms. Int J Perceptive Cogn Comput 6(2):97–106. https://doi.org/10.31436/ijpcc.v6i2.165
16. Khan R et al (Aug.2021) Machine learning and IoT-based waste management model. Comput Intell Neurosci 2021:1–11. https://doi.org/10.1155/2021/5942574
17. Hanbal IF, Ingosan JS, Oyam NAA, Hu Y (Apr.2020) Classifying wastes using random forests, Gaussian Naïve Bayes, support vector machine and multilayer perceptron. IOP Conf Ser Mater Sci Eng 803(1):012017. https://doi.org/10.1088/1757-899X/803/1/012017
18. Costa B. S. *et al.* Artificial intelligence in automated sorting in trash recycling, in *Anais do XV Encontro Nacional de Inteligência Artificial e Computacional (ENIAC 2018)*, Oct. 2018, no. August, pp. 198–205. https://doi.org/10.5753/eniac.2018.4416.
19. Hameed K., Ahmed M., and Abed M. A manifold intelligent decision system for fusion and benchmarking of deep waste-sorting models, Eng Appl Artif Intell. vol. 132, no. January, p. 107926, 2024, https://doi.org/10.1016/j.engappai.2024.107926.
20. M. G. C. P, S. Yadav, A. Shanmugam, H. V, and N. Suresh, Waste classification and segregation: machine learning and IOT approach, in *2021 2nd International Conference on Intelligent Engineering and Management (ICIEM)*, Apr. 2021, pp. 233–238. https://doi.org/10.1109/ICIEM51511.2021.9445289.
21. Sakr G. E., Mokbel M., Darwich A., Khneisser M. N., and Hadi A. Comparing deep learning and support vector machines for autonomous waste sorting, in *2016 IEEE International Multidisciplinary Conference on Engineering Technology (IMCET)*, Nov. 2016, pp. 207–212. https://doi.org/10.1109/IMCET.2016.7777453.
22. Masand A, Chauhan S, Jangid M, Kumar R, Roy S (2021) ScrapNet: an efficient approach to trash classification. IEEE Access 9:130947–130958. https://doi.org/10.1109/ACCESS.2021.3111230
23. Bohong L. and Xinpeng W. Garbage detection algorithm based on YOLO v3, in *2022 IEEE International Conference on Electrical Engineering, Big Data and Algorithms (EEBDA)*, Feb. 2022, pp. 784–788. https://doi.org/10.1109/EEBDA53927.2022.9744738.
24. Pan Z. Research on improved Yolo on garbage classification task, in *2022 IEEE International Conference on Electrical Engineering, Big Data and Algorithms (EEBDA)*, Feb. 2022, pp. 951–953. https://doi.org/10.1109/EEBDA53927.2022.9744865.
25. Aishwarya A., Wadhwa P., Owais O., and Vashisht V. A waste management technique to detect and separate non-biodegradable waste using machine learning and YOLO algorithm, in *2021 11th International Conference on Cloud Computing, Data Science & Engineering (Confluence)*, Jan. 2021, pp. 443–447. https://doi.org/10.1109/Confluence51648.2021.9377163.
26. De Carolis B., Ladogana F., and Macchiarulo N. YOLO TrashNet: garbage detection in video streams, in *2020 IEEE Conference on Evolving and Adaptive Intelligent Systems (EAIS)*, May 2020, pp. 1–7. https://doi.org/10.1109/EAIS48028.2020.9122693.
27. Tornero P., Puente S., and Gil P. Detection and location of domestic waste for planning its collection using an autonomous robot, in *2022 8th International Conference on Control, Automation and Robotics (ICCAR)*, Apr. 2022, pp. 138–144. https://doi.org/10.1109/ICCAR55106.2022.9782609.
28. Lin W. YOLO-Green: a real-time classification and object detection model optimized for waste management, in *2021 IEEE International Conference on Big Data (Big Data)*, Dec. 2021, pp. 51–57. https://doi.org/10.1109/BigData52589.2021.9671821.
29. Buragohain A, Mali B, Saha S, Singh PK (May2022) A deep transfer learning based approach to detect COVID-19 waste. Internet Technol Lett 5(3):114–119. https://doi.org/10.1002/itl2.327
30. Middya A. I., Chattopadhyay D, and Roy S. Garbage detection and classification using Faster-RCNN with Inception-V2, in *2021 IEEE 18th India Council International Conference (INDICON)*, Dec. 2021, pp. 1–6. https://doi.org/10.1109/INDICON52576.2021.9691547.
31. Kumar NM et al (Aug.2021) Artificial intelligence-based solution for sorting COVID related medical waste streams and supporting data-driven decisions for smart circular economy practice. Process Saf Environ Prot 152(July):482–494. https://doi.org/10.1016/j.psep.2021.06.026
32. Hermawan I., Mardiyono A., Iswara R. W., Murad F. A., Ardiawan M. A., and Puspita R. Development of Covid medical waste object classification system using YOLOv5 on Raspberry Pi, *2023 10th Int. Conf. Inf. Technol. Comput. Electr. Eng. ICITACEE 2023*, pp. 443–447, 2023, https://doi.org/10.1109/ICITACEE58587.2023.10277207.
33. Ahmed Z and Sadat Shanto S. Performance analysis of YOLO architectures for surgical waste detection in post-COVID-19 medical waste management. Malaysian J Sci Adv Technol. vol. 4, no. 1, pp. 1–9, Jan. 2024, https://doi.org/10.56532/mjsat.v4i1.232.
34. Wu B., Pang C., Zeng X., and Hu X. ME-YOLO: improved YOLOv5 for detecting medical personal protective equipment. Appl Sci. vol. 12, no. 23, 2022, https://doi.org/10.3390/app122311978.
35. Ferdous M, Ahsan SMM (2022) A computer vision-based system for surgical waste detection. Int J Adv Comput Sci Appl 13(3):554–565. https://doi.org/10.14569/IJACSA.2022.0130366

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